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ORIGINAL ARTICLE

Disturbance Observer-Based On-Policy Adaptive Dynamic Programming for a Quadrotor UAV with External Disturbance

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Abstract
This work is an investment in addressing the optimal trajectory tracking control problem for a quadrotor unmanned aerial vehicle (UAV) subjected to external disturbance. The proposed method is a combination of an Adaptive Dynamic Programming (ADP)-based controller and a nonlinear disturbance observer (NDO), ensuring the stability of the closed-loop quadrotor under the influence of unknown external disturbance and achieving near-optimal trajectory tracking control. Firstly, the tracking error dynamics are established by utilizing the feedforward terms to compensate for the known time-varying error information in the quadrotor dynamics. Then, a nonlinear disturbance observer (NDO) is designed to compensate for unknown external disturbances. By eliminating the influence of external disturbances and time-varying dynamics in the quadrotor system, the error model of the quadrotor is transferred to the nominal model. A critic-only learning structure is applied for both position and attitude controllers to learn the optimal feedback control policy. Finally, a flight simulation is given to illustrate the superior efficiency of the proposed control scheme. The simulation result shows that the proposed method lies in its ability to compute a smooth optimal control policy even under the influence of disturbances, while effectively mitigating the chattering effect.

Keywords Quadrotor · Adaptive dynamic programming · Critic-only learning · Disturbance observer

1 Introduction

In recent years, unmanned aerial vehicles (UAVs) have received a lot of attention from academic and industrial fields due to their wide applications, such as environmental mapping [1], logistics delivery [2], search and rescue operations [3], military surveillance [4], and some other applications. In practice, quadrotor UAVs are the most popular due to many significant advantages, including vertical take-off and landing, simple structure, lightweight, low manufacturing cost, and hovering ability [5].

For a quadrotor UAV system, tracking control design is one of the key issues. However, the problem of control design encounters some challenges due to the high nonlinear and strong coupling. Moreover, the quadrotor system is an underactuated system, as it has six degrees of freedom (DoF) but only four independent inputs, which are regulated by adjusting the speeds of its four actuators. Furthermore, external disturbances and model uncertainties should be considered during flight, providing more practical value in combat applications. To address these problems, numerous control strategies have been proposed, such as backstepping control [6], sliding mode control (SMC) [7], active disturbance rejection control (ADRC) [8], fuzzy logic control (FL) [9], and neural network (NN)-based control [10]. For a review of the literature, although these nonlinear control approaches achieve successful tracking performance and guarantee the stability of the closed-loop quadrotor system based on the Lyapunov stability theory. However, the problem of optimal tracking control was not discussed, resulting in unnecessary energy loss. Due to the limited battery

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capacity of quadrotor UAVs, optimal tracking control has been used to overcome this limitation and improve their performance and efficiency.

In the control systems engineering community, optimal control has always attracted significant attention and remains a highly influential research topic [11–13]. Theoretically, the optimal control problem typically involves solving the Algebraic Riccati Equation (ARE) for linear systems [14] and the Hamilton-Jacobi-Bellman (HJB) equation for nonlinear systems [15]. However, it is more difficult or impossible to find the optimal solution of the HJB equation for highly nonlinear systems such as quadrotor UAVs using only mathematical analysis [16]. To overcome this challenge, in recent years, reinforcement learning (RL)-based control approaches have provided more considerable advantages to solve optimal control problems, which find the optimal solution without directly solving the HJB equation. The adaptive dynamic programming (ADP) technique is a class of RL that serves as a bridge between traditional optimal control methods and adaptive control algorithms. To the best of our knowledge, ADP techniques utilize neural networks (NNs) as a feasible approach. The traditional ADP structure is the actor-critic (AC) structure, which was developed by Lewis et al. for nonlinear systems to solve optimal regulation control [17] and optimal tracking control [18]. According to a predefined cost function, the critic NN is employed to approximate the cost function, while the actor NN is employed to approximate an improving control policy. However, training two NNs typically requires significant computational resources, making it challenging to implement the algorithm in practical applications. To overcome this limitation, Ouyang et al. [19] developed a critic-only learning structure that was superior to the AC structure in computation reduction when using only a NN. For both learning structures, a necessary condition called persistent excitation (PE) condition must be satisfied to guarantee the existence of weight update laws for NNs. However, the ADP technique performs poorly in handling external time-varying disturbances, thus failing to meet the performance requirements of disturbance attenuation when quadrotor UAVs fly in complex environments. As a result, this indicates that the high-quality control performance of practical quadrotor systems cannot be achieved only by the ADP technique.

The quadrotor UAV usually suffers the influence of unknown external disturbances from aerospace effects, including aerodynamic drag and lift forces, as well as friction forces [20]. As mentioned earlier, the ADP-based control is not able to deal with time-varying external disturbances. Therefore, it usually requires rejecting the influence of external disturbance before implementing the ADP algorithm. The SMC can effectively robustly respond to external disturbances that satisfy bounded conditions. Zhao et al. [21] developed an SMC-based ADP controller to obtain approximate optimal control for a class of uncertain nonlinear systems. However, as a common problem with SMC methods, chattering reduction must be considered. In contrast, the disturbance observer (DO) method is considered to estimate and compensate for unknown external disturbances [22]. Under the assumption of a small-bounded disturbance change rate, the estimated disturbance error converges to a small-bounded neighborhood of zero and is set to zero. Duo et al. [23] introduced a robust controller constructed by an ADP controller and an adaptive disturbance observer (ADO). There, an ADO is employed to estimate the external disturbance with an unknown upper bound, and the ADP-based controller is utilized to drive the tracking errors of the quadrotor UAV to zero. However, the problem of considering the position and attitude dynamics as a nominal linear model subject to disturbance may result in a large upper-bounded disturbance, leading to significant estimation errors.

Inspired by previous works [23] and these previous discussions, this paper proposes an ADP-based DO controller for the nonlinear quadrotor UAV subjected to unknown external time-varying disturbances. The main contribution is listed as follows:

1. Compared to [23], the approach implements the ADP-based control for the quadrotor system to simplify the quadrotor dynamics as a nominal linear system subjected to disturbance. The proposed ADP-based control adopts a non-linear quadrotor dynamic model to learn the near-optimal solution.
2. The NDO is designed to eliminate the influence of unknown external time-varying disturbances in the quadrotor dynamics. Under the assumption of a small disturbance change rate, the external disturbance can be estimated and compensated accurately, and the estimated disturbance error converges to a small bounded neighborhood of zero and is set to zero.
3. The updated NN weight for critic-only learning is designed to learn the near-optimal solution online. The proposed ADP-based NDO controller achieves approximate optimal trajectory tracking and guarantees the stability of the quadrotor system under the influence of external disturbances.

The rest of this paper is organized as follows. In Sect. 2, the mathematical model and model transformation of the quadrotor are presented. Section 3 introduces the ADP-based NDO algorithm. Section 4, the simulation results and analysis are provided. Finally, Sect. 5 concludes this paper

Notation: Let $I_n \in \mathbb{R}^{n\times n}$ be the identity matrix, $0_{n\times m} \in \mathbb{R}^{n\times m}$ be a zero matrix, and $c_{i,j} \in \mathbb{R}^{i\times 1}$ be a vector with 1 in the element j^{th} and 0 otherwise.

2 Quadrotor System

2.1 Mathematical Model

The quadrotor system is an underactuated system with six degrees of freedom (DoF) but only four independent control inputs provided by its four rotors. Define the Earth-fixed frame $\mathbf{E} = \{\mathbf{x_e y_e z_e}\}$ and the body-fixed frame $\mathbf{B} = \{\mathbf{x_b y_b z_b}\}$ as shown in Fig. 1.

Let's define $p = [x, y, z]^T \in \mathbb{R}^3$ as the position of the quadrotor mass center in the Earth-fixed frame and $\eta = [\phi, \theta, \psi]^T \in \mathbb{R}^3$ as Euler angles, where ϕ, θ, and ψ represent the roll, pitch, and yaw angles, respectively. The roll and pitch angles are bounded satisfying $|\phi| \le \pi/2$ and $|\theta| \le \pi/2$ to avoid singularity issues in Euler angles. The rotation matrix which transfers the coordinates of the quadrotor from the body-fixed frame to the Earth-fixed frame can be obtained as.

$$R_t(\eta) = \begin{bmatrix} c_\psi c_\theta & c_\psi s_\theta s_\phi - s_\psi c_\phi & c_\psi s_\theta c_\phi + s_\psi s_\phi \\ s_\psi c_\theta & s_\psi s_\theta s_\phi + c_\psi c_\phi & s_\psi s_\theta c_\phi - c_\psi s_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \tag{1}$$

where s_α and c_α with $\alpha = \phi, \theta, \psi$ represent $\sin(\alpha)$ and $\cos(\alpha)$ functions, respectively. Define the command control inputs u_z and $u_\eta = [u_\phi, u_\theta, u_\psi]^T$ as

$$\begin{aligned} &u_z = k_z \sum_{i=1}^{4} \Omega_i^2, \quad u_\phi = lk_z(\Omega_2^2 - \Omega_4^2), \quad u_\theta = lk_z(\Omega_1^2 - \Omega_3^2) \\ &u_\psi = k_\tau \sum_{i=1}^{4} (-1)^{i+1} \Omega_i^2 \end{aligned} \tag{2}$$

in where Ω_i is the speed of the ith rotor, l is the distance from the quadrotor mass center to rotors, k_z is the thrust coefficient, and k_τ is the drag coefficient. As described in [24, 25], the dynamics of a quadrotor consist of two interconnected systems: position and attitude dynamics, which are coupled through the rotation matrix as

$$\begin{aligned} m\ddot{p} &= -mgc_{3,3} + R_t(\eta)u_z c_{3,3} + d_p \\ J\ddot{\eta} &= C(\eta, \dot{\eta})\dot{\eta} + u_\eta + d_\eta \end{aligned} \tag{3}$$

where m is the mass of the quadrotor, g is the gravitational acceleration, and $J = \mathrm{diag}([J_x, J_y, J_z]) \in \mathbb{R}^{3\times 3}$

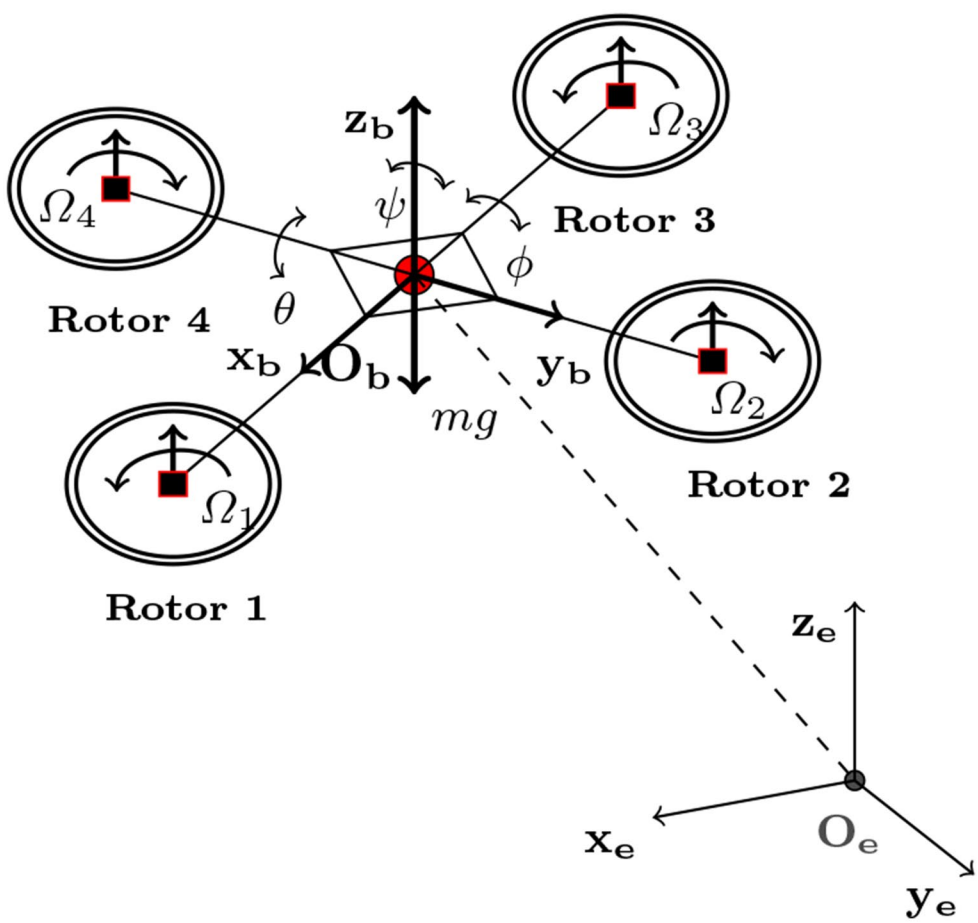

Fig. 1 The quadrotor coordinate frame

is the inertia matrix. $d_p = [d_x, d_y, d_z]^T \in \mathbb{R}^3$ and $d_\eta = [d_\phi, d_\theta, d_\psi]^T \in \mathbb{R}^3$ are external disturbance. According to [26], the Coriolis term $C(\eta, \dot{\eta}) \in \mathbb{R}^{3\times 3}$ is described as

$$C(\eta, \dot{\eta}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \tag{4}$$

where

$$\begin{aligned}
c_{11} =& 0 \\
c_{12} =& (J_y - J_z)(\dot{\theta} c_\phi s_\phi + \dot{\psi} s_\phi^2 c_\theta) - J_x \dot{\psi} c_\theta \\
&+ (J_z - J_y)\dot{\psi} c_\phi^2 c_\theta \\
c_{13} =& (J_z - J_y)\dot{\psi} c_\phi s_\phi c_\theta^2 \\
c_{21} =& (J_z - J_y)(\dot{\theta} c_\phi s_\phi + \dot{\psi} s_\phi^2 c_\theta) + J_x \dot{\psi} c_\theta \\
&+ (J_y - J_z)\dot{\psi} c_\phi^2 c_\theta \\
c_{22} =& (J_z - J_y)\dot{\phi} c_\phi s_\phi \\
c_{23} =& - J_x \dot{\psi} s_\theta c_\theta + J_z \dot{\psi} c_\phi^2 c_\theta s_\theta + J_y \dot{\psi} s_\theta^2 c_\theta s_\theta \\
c_{31} =& (J_y - J_z)\dot{\psi} c_\theta^2 c_\phi s_\phi - J_x \dot{\theta} c_\theta \\
c_{32} =& (J_z - J_y)(\dot{\theta} c_\phi s_\phi s_\theta + \dot{\phi} s_\phi^2 c_\theta) + (J_y - J_z)\dot{\phi} c_\phi^2 c_\theta \\
&+ J_x \dot{\psi} s_\theta c_\theta - J_y \dot{\psi} s_\phi^2 c_\theta s_\theta - J_z \dot{\psi} c_\phi^2 c_\theta s_\theta \\
c_{33} =& - J_z \dot{\theta} c_\phi^2 s_\theta c_\theta - J_y \dot{\theta} s_\phi^2 c_\theta s_\theta + J_x \dot{\theta} s_\theta c_\theta \\
&+ (J_y - J_z)\dot{\phi} c_\phi s_\phi c_\theta^2
\end{aligned} \tag{5}$$

Remark 1 It is easy to observe that the dynamic model of the quadrotor system subjected to external disturbance is strong coupling and underactuated via the rotation matrix due to six degrees of freedom (DoF) and only four independent inputs.

Assumption 1 *Due to the bound of physical condition, the external disturbances and their derivatives are assumed to be bounded as*

$$\begin{aligned}
&||d_p|| \le D_{p1}, \quad ||d_\eta|| \le D_{\eta 1} \\
&||\dot{d}_p|| \le D_{p2}, \quad ||\dot{d}_\eta|| \le D_{\eta 2}
\end{aligned} \tag{6}$$

2.2 Model Transformation

To address the underactuated system, a hierarchical control structure consisting of a position controller and an attitude controller is proposed as shown in Fig. 2, and a virtual position control input $u_p = [u_{p,x}, u_{p,y}, u_{p,z}]^T$ is defined as follows

$$\begin{aligned}
u_{p,x} &= u_z\left(c_\psi s_\theta c_\phi + s_\psi s_\phi\right) \\
u_{p,y} &= u_z\left(s_\psi s_\theta c_\phi - c_\psi s_\phi\right) \\
u_{p,z} &= u_z c_\phi c_\theta
\end{aligned} \tag{7}$$

Similar to [27], once the virtual position control input is determined, the desired Euler angles $\eta_d = [\phi_d, \theta_d, \psi_d]^T$ can be obtained by setting the desired yaw angles ψ_d to constant.

$$\begin{aligned}
u_z &= \sqrt{u_{p,x}^2 + u_{p,y}^2 + u_{p,z}^2} \\
\phi_d &= \arcsin\left(\frac{u_{p,x}\sin\psi_d - u_{p,y}\cos\psi_d}{u_z}\right) \\
\theta_d &= \arctan\left(\frac{u_{p,x}\cos\psi_d + u_{p,y}\sin\psi_d}{u_{p,z}}\right)
\end{aligned} \tag{8}$$

Let the desired references for position and attitude subsystems as $p_d = [x_d, y_d, z_d]^T \in \mathbb{R}^3$ and $\eta_d = [\phi_d, \theta_d, \psi_d]^T \in \mathbb{R}^3$. To design the trajectory tracking controller for the quadrotor, the position and attitude tracking errors are defined as

$$\begin{aligned}
&\begin{cases} e_{p1} &= p - p_d = [e_x, e_y, e_z]^T \\ e_{p2} &= \dot{p} - \dot{p}_d = [\dot{e}_x, \dot{e}_y, \dot{e}_z]^T \end{cases} \\
&\begin{cases} e_{\eta 1} &= \eta - \eta_d = [e_\phi, e_\theta, e_\psi]^T \\ e_{\eta 2} &= \dot{\eta} - \dot{\eta}_d = [\dot{e}_\phi, \dot{e}_\theta, \dot{e}_\psi]^T \end{cases}
\end{aligned} \tag{9}$$

Hence, the position and attitude error subsystems are established as follows

$$\begin{cases} \dot{e}_{p1} &= e_{p2} \\ \dot{e}_{p2} &= -g c_{3,3} + m^{-1} u_p + m^{-1} d_p - \ddot{p}_d \end{cases} \tag{10}$$

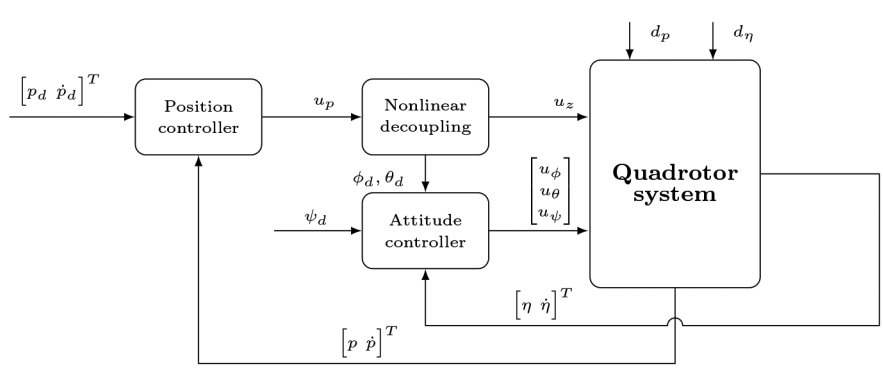

Fig. 2 The hierarchical control structure of the quadrotor

$$\begin{cases}\dot{e}_{\eta 1} &= e_{\eta 2}\\ \dot{e}_{\eta 2} &= J^{-1}C(\eta,\dot{\eta})\dot{\eta} + J^{-1}u_\eta + J^{-1}d_\eta - \ddot{\eta}_d\end{cases} \tag{11}$$

The position and attitude control inputs are divided into two parts as

$$\begin{aligned} u_p &= u_p^o + u_p^f\\ u_\eta &= u_\eta^o + u_\eta^f \end{aligned} \tag{12}$$

The feedforward terms $u_p^f = mgc_{3,3} + m\ddot{p}_d$ and $u_\eta^f = C(\eta,\dot{\eta})\dot{\eta}_d + J\ddot{\eta}_d$ is designed to compensate for time-varying known information of the desired reference and guarantees perfect tracking. Meanwhile, the feedback terms u_p^o and u_η^o are designed to guarantee the stability of the position and attitude error dynamics (10) & (11) in an optimal manner by minimizing the performance index

$$V_i = \int_t^\infty (z_i^T Q_i z_i + [u_i^o]^T R_i u_i^o) d\tau \tag{13}$$

where $z_i = [e_{i1}^T, e_{i2}^T]^T \in \mathbb{R}^6$, and $Q_i \in \mathbb{R}^{6\times 6}$ and $R_i \in \mathbb{R}^{3\times 3}$ are positive define matrices with $i = p, \eta$. By solving the HJB equation respected to the performance index (13), the optimal feedback terms $u_p^{o,*}$ and $u_\eta^{o,*}$ can be obtained.
Applying the control input laws (12) to the position and attitude error dynamics in (10) & (11), the errors dynamics can be obtained as

$$\begin{cases}\dot{e}_{p1} &= e_{p2}\\ \dot{e}_{p2} &= m^{-1}u_p^o + m^{-1}d_p\end{cases} \tag{14}$$

$$\begin{cases}\dot{e}_{\eta 1} &= e_{\eta 2}\\ \dot{e}_{\eta 2} &= J^{-1}C(\eta,\dot{\eta})e_{\eta 2} + J^{-1}u_\eta^o + J^{-1}d_\eta\end{cases} \tag{15}$$

For simplification, error dynamic subsystems (14) & (15) can be rewritten as a common form as

$$\begin{aligned} \dot{e}_{i1} &= e_{i2}\\ \dot{e}_{i2} &= f_i(e_{i1}, e_{i2}) + g_i u_i^o + \Delta_i \end{aligned} \tag{16}$$

The function $f_i(e_{i1}, e_{i1})$, g_i, and Δ_i represent

$$\begin{aligned} f_p(e_{p1}, e_{p2}) &= 0_{3\times 1}\\ f_\eta(e_{\eta 1}, e_{\eta 2}) &= J^{-1}C(\eta,\dot{\eta})e_{\eta 2}\\ g_p &= m^{-1}I_3\\ g_\eta &= J^{-1}I_3\\ \Delta_p &= m^{-1}d_p\\ \Delta_\eta &= J^{-1}d_\eta \end{aligned} \tag{17}$$

It is noted that the control matrices g_i with $i = p, \eta$ are constant matrices. By compensating for disturbances Δ_i, the main objective of this paper is transferred to finding the optimal feedback control laws u_p^o and u_η^o to guarantee the stability of the subsystems (16) and achieve the optimal trajectory tracking through the performance index in (13).

3 Main Results

3.1 Disturbance Observer Design

For the subsystems (16), a nonlinear disturbance observer is proposed to estimate disturbances Δ_i in the position and attitude subsystems.

$$\begin{aligned} \hat{\Delta}_i &= x_i + r_i(e_{i2})\\ \dot{x}_i &= -\frac{\partial r_i(z_i)}{\partial z_i}\Big(x_i + r_i(e_{i2}) + f_i + g_i u_i^o\Big) \end{aligned} \tag{18}$$

where $x_i \in \mathbb{R}^3$ is the internal states of the nonlinear disturbance observer, $\hat{\Delta}_i$ is the estimated disturbance of Δ_i, and $r_i(e_{i2})$ is a nonlinear vector-valued function.

Assumption 2 *The disturbances in the position and attitude subsystems are assumed to be unknown and slowly time-varying. It means that the derivative of disturbances D_{p2} and $D_{\eta 2}$ are small.*

Theorem 1 For the system (12) and the nonlinear disturbance observer in (18), the disturbance estimation error $\tilde{\Delta}_i = \Delta_i - \hat{\Delta}_i$ is ultimately uniformly bounded (UUB) stable.

Proof Choosing the Lyapunov function as

$$L_{1i} = \frac{1}{2}\tilde{\Delta}_i^T\tilde{\Delta}_i \tag{19}$$

Taking the time derivative of V_i along the trajectory (12) as

$$\begin{aligned} \dot{L}_{1i} =& \tilde{\Delta}_i^T\dot{\tilde{\Delta}}_i\\ =& -\tilde{\Delta}_i^T\big(\dot{x}_i + \dot{r}_i(e_{i2})\big) + \tilde{\Delta}_i^T\dot{\Delta}_i\\ \le& -\tilde{\Delta}_i^T\Big(-\frac{\partial r_i(e_{i2})}{\partial e_{i2}}\Big(x_i + r_i(e_{i2}) + f_i + g_i u_i^o\Big)\\ &+ \frac{\partial r_i(e_{i2})}{\partial e_{i2}}\dot{e}_{i2}\Big) + \frac{1}{2}\tilde{\Delta}_i^T\tilde{\Delta}_i + \frac{1}{2}\dot{\Delta}_i^T\dot{\Delta}_i\\ =& -\tilde{\Delta}_i^T\Big(-\frac{\partial r_i(e_{i2})}{\partial e_{i2}}\Big(\hat{\Delta}_i + f_i + g_i u_i^o\Big) + \frac{1}{2}\tilde{\Delta}_i^T\tilde{\Delta}_i\\ &+ \frac{\partial r_i(e_{i2})}{\partial e_{i2}}\Big(f_i + g_i u_i^o + \Delta_i\Big)\Big) + \frac{1}{2}\dot{\Delta}_i^T\dot{\Delta}_i\\ \le& -\tilde{\Delta}_i^T\Big(\frac{\partial r_i(e_{i2})}{\partial e_{i2}} - \frac{1}{2}\Big)\tilde{\Delta}_i + \frac{1}{2}\dot{\Delta}_i^T\dot{\Delta}_i \end{aligned} \tag{20}$$

Let $\Lambda_i = \partial r_i(e_{i2})/\partial e_{i2}$ and $\epsilon_i = 0.5\dot{\Delta}_i^T\dot{\Delta}_i$. Hence, we obtain

$$\dot{L}_{1i} \leq -2\Big(\lambda_{\min}(\Lambda_i) - 0.5\Big)L_{1i} + \epsilon_i \tag{21}$$

where $\lambda_{\min}(\Lambda_i)$ is the minimum eigenvalue of Λ_i. Because g_i is bounded, $|g_i| \leq \bar{g}_i$, the positive part ϵ_i is also bounded as

$$\epsilon_i \leq 0.5D_{2i}^T\bar{g}_i^T\bar{g}_iD_{2i} \tag{22}$$

By choosing Λ_i as a positive define matrix satisfying $\lambda_{\min}(\Lambda_i) > 0.5$, the estimated disturbance error $\tilde{\Delta}_i$ will converge to a neighborhood of zero and be set to zero.

3.2 ADP-Based Control Design

By compensating for disturbances using disturbance observer in this previous section, the closed-loop of the error position and attitude dynamics can be obtained as

$$\begin{aligned}\dot{e}_{i1} &= e_{i2}\\ \dot{e}_{i2} &= f_i(e_{i1}, e_{i2}) + g_iu_i^o + \tilde{\Delta}_i\end{aligned} \tag{23}$$

It is noted that the estimated disturbance error $\tilde{\Delta}_i$ will converge to a neighborhood of zero and be set to zero. Therefore, this session aims to design an ADP-based control policy to achieve the optimal feedback terms u_p^o and u_η^o, which is considered by eliminating the uncertain term $\tilde{\Delta}_i$. Let's define the position and attitude states $z_i = [z_{i1}^T, z_{i2}^T]^T \in \mathbb{R}^6$ with $i = p, \eta$. The position and attitude subsystems are rewritten as

$$z_i = F_i(z_i) + G_iu_i^o \tag{24}$$

where $i = p, \eta$ and $F_i(z_i) = [e_{i2}^T, f_i(e_{i1}, e_{i2})^T]^T \in \mathbb{R}^6$, $G_i = [0_{3\times3}, g_i]^T$. Recall the performance index (13) as

$$V_i(z_i) = \int_t^\infty r(z_i, u_i^o)d\tau \tag{25}$$

where $r(z_i, u_i^0) = z_i^TQ_iz_i + [u_i^o]^TR_iu_i^o$. In general, the optimal feedback control policy $u_i^{o,*}$ for implementing the minimization of the performance index $V_i(z_i)$ respected to the system (24). The Hamilton function is given as

$$H_i(z_i, u_i^o, \nabla V_i) = r_i\big(z_i, u_i^o\big) + [\nabla V_i^*]^T\Big(F_i(z_i) + G_iu_i^o\Big) \tag{26}$$

where ∇V_i is the partial derivative of the discounted performance index $V_i(z_i)$ with respect to z_i and $V_i^*(z_i)$ is the optimal discounted performance index defined as

$$V_i^*(z_i) = \min_{u_i^o}\int_t^\infty r_i\big(z_i, u_i^o\big)d\tau \tag{27}$$

Under the optimality condition $\partial H_i(z_i, u_i^o, \nabla V_i^*)/\partial z_i = 0$, the optimal control policy $u_i^{o,*}$ can be derived as follows

$$u_i^{o,*} = -\frac{1}{2}R_i^{-1}G_i^T\nabla V_i^*(z_i) \tag{28}$$

Substituting $u_i^{o,*}$ into (26), the HJB equation can be rewritten as

$$\begin{aligned}H_i^*(z_i, u_i^{o,*}, \nabla V_i^*) =& z_i^TQ_iz_i + [\nabla V_i^*]^TF_i(z_i)\\ &- \frac{1}{4}[\nabla V_i^*]^TG_iR_i^{-1}G_i^T\nabla V_i^* = 0\end{aligned} \tag{29}$$

It is widely acknowledged that it is impossible to analytically solve the HJB equation for nonlinear systems. Therefore, this paper proposes a critic-only design to achieve near-optimal control, where the Neural Network (NN) is utilized to approximate the performance index as

$$V_i^*(z_i) = W_i^T\varphi_i(z_i) + \varepsilon_i(z_i) \tag{30}$$

where $W_i = [W_{i1}, ..., W_{il_i}]^T \in \mathbb{R}^{l_i}$ is the ideal critic weight, l_i as the number of neurons in the hidden layer, $\varphi_i(z_i) = [\varphi_{i1}(z_i), ...,\varphi_{il_i}(z_i)]^T : \mathbb{R}^6 \to \mathbb{R}^{l_i}$ is the action function satisfying $\varphi_i(0_{6\times1}) = 0_{l_1\times1}$ and $\nabla\varphi_i(0_{6\times1}) = 0_{l_1\times1}$, and $\varepsilon_i(z_i)$ is the reconstruction error.

Assumption 3 *The ideal critic weight W_i and the reconstruction error $\varepsilon_i(z_i)$ are assumed to be bounded satisfying $||W_i|| \leq \bar{W}_i$ and $||\varepsilon_i(z_i)|| \leq \bar{\varepsilon}_i$ with $\bar{W}_i$ and $\bar{\varepsilon}_i$ are positive contacts. Additionally, the gradient of the reconstruction error respected to z_i satisfies $||\nabla\varepsilon_i(z_i)|| \leq \nabla\bar{\varepsilon}_i$ with $\nabla\bar{\varepsilon}_i$ is a positive constant.*

Thus, the optimal feedback control input can be rewritten as

$$u_i^{o,*} = -\frac{1}{2}R_i^{-1}G_i^T\Big(\nabla\varphi_i(z_i)^TW_i + \nabla\varepsilon_i(z_i)\Big) \tag{31}$$

where $\nabla\varphi_i(z_i) = \partial\varphi_i(z_i)/\partial z_i$. It is important to clarify the approximation error for the HJB Eq. (29) with the function approximation (30)

$$\begin{aligned}H_i(z_i, W_i) =& z_i^TQ_iz_i + W_i^T\nabla\varphi_iF_i\\ &- \frac{1}{4}W_i^T\nabla\varphi_iG_iR_i^{-1}G_i^T\nabla\varphi_i^TW_i = \epsilon_{r,i}\end{aligned} \tag{32}$$

where $\epsilon_{hjb,i}$ is the residual error

$$\epsilon_{r,i} = -\nabla\varepsilon_i F_i + \frac{1}{2}W_i^T\nabla\varphi_i G_i R_i^{-1} G_i^T \nabla\varepsilon_i + \frac{1}{4}\nabla\varepsilon_i^T G_i R_i^{-1} G_i^T \nabla\varepsilon_i \tag{33}$$

Assumption 4 *For the system (24), this is assumed that there exists $\xi(z_i) \geq 0$ such that $||F_i(z_i) + G_i u_i^{o,*}|| \leq \xi(z_i)$ with $\forall z_i$. There is a Lyapunov function $J_i(z_i)$ satisfying $J_i(0) = 0$ and $\dot{J}_i(z_i) = \nabla J_i^T(z_i)\dot{z}_i = [\nabla J_i(z_i)]^T(F_i(z_i) + G_i u_i^{o,*}) < 0$. Let $\xi(z_i) = \kappa_i||\nabla J_i^T(z_i)||$, then $\dot{J}_i(z_i) = -\nabla J_i^T(z_i)\Pi_i\nabla J_i^T(z_i) \leq 0$ with Π_i is positive define matrix.*

Remark 2 *Under the **Assumption** 4, the residual error $\epsilon_{r,i}$ is also bounded satisfying $||\epsilon_{r,i}|| \leq \bar{\epsilon}_{r,i}$ with $\bar{\epsilon}_{r,i}$ is a positive constant. Furthermore, the residual error $\epsilon_{r,i}$ uniformly converges to zero as the number of neurons in the hidden layer l_i is large enough [17].*

Due to the unknown of the ideal critic weight, the approximation output of the critic NN is

$$\hat{V}_i(z_i) = \hat{W}_i^T\varphi_i(z_i) \tag{34}$$

where $\hat{W}_i$ is the current estimated value of the ideal critic weight and $\hat{V}_i$ is the estimated value of the performance index. Usually, the optimal feedback control policy can also be approximated as

$$\hat{u}_i^o = -\frac{1}{2}R_i^{-1}G_i^T\nabla\varphi_i(z_i)^T\hat{W}_i \tag{35}$$

Thus, the approximated HJB equation can be derived as

$$H_i(z_i, \hat{W}_i) = z_i^T Q_i z_i + \hat{W}_i^T\nabla\varphi_i F_i - \frac{1}{4}\hat{W}_i^T\nabla\varphi_i G_i R_i^{-1} G_i^T \nabla\varphi_i^T \hat{W}_i = e_{H,i} \tag{36}$$

where $e_{H,i}$ is the HJB error

$$e_{H,i} = -\tilde{W}_i^T\gamma_i(z_i, \hat{u}_i^o) + \frac{1}{4}\tilde{W}_i^T\nabla\varphi_i G_i R_i^{-1} G_i^T \nabla\varphi_i^T \tilde{W}_i + \epsilon_{r,i} \tag{37}$$

where $\gamma_i(z_i, \hat{u}_i^o) = \nabla\varphi(z_i)\Big(F(z_i) + G_i\hat{u}_i^o\Big)$ and $\tilde{W}_i = W_i - \hat{W}_i$is the estimated weight error. When $\hat{W}_i(t) \to W_i, e_{H,i} \to \epsilon_{r,i}$. The problem is to find an updated weight law to adjust the critic weight so that the HJB error $e_{H,i}$ is minimized. Consider the objective function as

$$E_{H,i} = \frac{1}{2}e_{H,i}^2 \tag{38}$$

According to [18], it is noted that the updated weight law by only minimizing the HJB error $e_{H,i}$ associated with the HJB Eq. (26) can not guarantee the stability of the closed-loop system (24). To address this problem, inspired by [19], the alternative updated weight law is modified as

$$\dot{\hat{W}}_i = \Gamma_{i1} + \Gamma_{i2} \tag{39}$$

with

$$\begin{aligned} \Gamma_{i1} &= -k_{i1}\frac{\gamma_i}{(1+\gamma_i^T\gamma_i)^2}e_{H,i} \\ \Gamma_{i2} &= \frac{1}{2}k_{i2}\zeta_i\nabla\varphi_i(z_i)G_i R_i^{-1} G_i^T z_i \end{aligned} \tag{40}$$

where k_{i1} and k_{i2} are positive constants, and

$$\zeta_i = \begin{cases} 0 & \text{if } z_i\Big(F(z_i) + G_i\hat{u}_i^o\Big) < 0 \\ 1 & \text{otherwise} \end{cases} \tag{41}$$

Based on the updated weight law in (39), the critic estimated weight error dynamics is given as

$$\begin{aligned} \dot{\tilde{W}}_i &= \dot{W}_i - \dot{\hat{W}}_i = -\dot{\hat{W}}_i \\ &= k_{i1}\frac{\gamma_i}{(1+\gamma_i^T\gamma_i)^2}e_{H,i} - \frac{1}{2}k_{i2}\zeta_i\nabla\varphi_i(z_i)\Xi_i z_i \\ &= -k_{i1}\frac{\gamma_i}{(1+\gamma_i^T\gamma_i)^2}\left(\tilde{W}_i^T\gamma_i - \frac{1}{4}\tilde{W}_i^T\Upsilon_i\tilde{W}_i - \epsilon_{r,i}\right) \\ &\quad - \frac{1}{2}k_{i2}\zeta_i\nabla\varphi_i\Xi_i z_i \end{aligned} \tag{42}$$

where $\Upsilon_i = \nabla\varphi_i\Xi_i\nabla\varphi_i^T \geq 0$ with $\Xi_i = G_i R_i^{-1} G_i^T \geq 0$ is a positive constant.

Remark 3 The first term of the updated weight law Γ_{i1} in (39) is designed to minimize the HJB error $e_{H,i}$ respected to the objective function $E_{B,i}$ in (38). The second term of the updated weight law Γ_{i2} in (39) is designed to ensure the stability of the closed-loop system based on the Lyapunov stability theory.

Assumption 5 *It is assumed that the term γ_i satisfies the persistence of excitation (PE) condition. There exist constants $\nu_{i1} > 0$, $\nu_{i2} > 0$, such that, for all t,*

$$\nu_{i1} I \leq \int_t^{t+T} \gamma_i(\tau)\gamma_i^T(\tau)d\tau \leq \nu_{i2} I \tag{43}$$

The PE condition is needed in adaptive control to identify the critic weights to approximate the performance index $V_i(z_i)$.

Theorem 2 Under the ***Assumption*** 5, the ADP-based controller with the updated weight law (39) for the system (24), $\tilde{W}_i$ and z_i are ultimate uniform bounded (UUB). In addition the near-optimal control objective is obtained with $||V_i^* - \hat{V}_i|| \leq \epsilon_{V_i}$ and $||u_i^{o,*} - \hat{u}_i^o|| \leq \epsilon_{u_i}$.

Proof: Choosing the Lyapunov candidate function as

$$L_{2i} = \frac{1}{2}\tilde{W}_i^T\tilde{W}_i + k_{i2}J_i(z_i) \tag{44}$$

Let $J_i(z_i) = (1/2)z_i^T z_i$, taking the time derivative of (44) as

$$\begin{aligned}\dot{L}_{2i} =& \tilde{W}_i^T\dot{W}_i + k_{i2}z_i^T\dot{z}_i \\ =& -k_{i1}\frac{\tilde{W}_i^T\gamma_i}{(1+\gamma_i^T\gamma_i)^2}\left(\tilde{W}_i^T\gamma_i - \frac{1}{4}\tilde{W}_i^T\Upsilon_i^T\tilde{W}_i - \epsilon_{r,i}\right) \\ &- \frac{1}{2}k_{i2}\zeta_i\tilde{W}_i^T\nabla\varphi_i\Xi_i z_i + k_{i2}z_i^T\left(F(z_i) + G_i\hat{u}_i^o\right)\end{aligned} \tag{45}$$

Case 1. For $\zeta_i = 0$ when $z_i^T\left(F(z_i) + G_i\hat{u}_i^o\right) < 0$, then we have

$$\dot{L}_{2i} < -k_{i1}\frac{\tilde{W}_i^T\gamma_i}{(1+\gamma_i^T\gamma_i)^2}\left(\tilde{W}_i^T\gamma_i - \frac{1}{4}\tilde{W}_i^T\Upsilon_i\tilde{W}_i - \epsilon_{r,i}\right) \tag{46}$$

Define $\sigma_i = 1/(1+\gamma_i\gamma_i^T)$. Then, we have

$$\begin{aligned}\dot{L}_{2i} \leq& -k_{i1}\sigma_i^2(\tilde{W}_i^T\gamma_i)^2 + \frac{1}{4}k_{i1}\sigma_i^2(\tilde{W}_i^T\gamma_i)\tilde{W}_i^T\Upsilon_i\tilde{W}_i \\ &+ k_{i1}\sigma_i^2(\tilde{W}_i^T\gamma_i)\epsilon_{r,i} \\ \leq& -k_{i1}\sigma_i^2||\tilde{W}_i^T\gamma_i||^2 + \frac{1}{2}k_{i1}\sigma_i^2\left(||\tilde{W}_i^T\gamma_i||^2 + ||\epsilon_{r,i}||^2\right) \\ &+ \frac{1}{8}k_{i1}\sigma_i^2\lambda_{\max}(\Upsilon_i)\left(||\tilde{W}_i^T\gamma_i||^2 + ||\tilde{W}_i||^4\right) \\ \leq& -\left(\frac{1}{2} - \frac{1}{8}\lambda_{\max}(\Upsilon_i)\right)k_{i1}\sigma_i^2||\tilde{W}_i||^2||\gamma_i||^2 \\ &+ \frac{1}{8}k_{i1}\sigma_i^2\lambda_{\max}(\Upsilon_i)||\tilde{W}_i||^4 + \frac{1}{2}k_{i1}\sigma_i^2||\epsilon_{r,i}||^2 \\ \leq& -k_{i1}\alpha_i||\tilde{W}_i||^4 + k_{i1}\varsigma_i\end{aligned} \tag{47}$$

where

$$\begin{aligned}\alpha_i &= \left(\frac{1}{4} - \frac{3}{16}\lambda_{\max}(\Upsilon_i)\right)\sigma_i^2 \\ \varsigma_i &= \frac{1}{2}\sigma_i^2||\epsilon_{r,i}||^2 - \left(\frac{1}{4} - \frac{1}{8}\lambda_{\max}(\Upsilon_i)\right)\sigma_i^2||\gamma_i||^4\end{aligned} \tag{48}$$

In order to $\dot{L}_{i2} \leq 0$m the following conditions should be satisfied

$$\begin{aligned}&\alpha_i > 0 \\ &||\tilde{W}|| > \sqrt[4]{\frac{\varsigma_i}{\alpha_i}}\end{aligned} \tag{49}$$

Case 2. $\zeta_i = 1$ when $z_i^T\left(F(z_i) + G_i\hat{u}_i^o\right) \geq 0$, then we have

$$\begin{aligned}\dot{L}_{2i} =& -k_{i1}\frac{\tilde{W}_i^T\gamma_i}{(1+\gamma_i^T\gamma_i)^2}\left(\tilde{W}_i^T\gamma_i - \frac{1}{4}\tilde{W}_i^T\Upsilon_i^T\tilde{W}_i - \epsilon_{r,i}\right) \\ &- \frac{1}{2}k_{i2}\tilde{W}_i^T\nabla\varphi_i\Xi_i z_i + k_{i2}z_i^T\left(F(z_i) + G_i\hat{u}_i^o\right) \\ =& -k_{i1}\frac{\tilde{W}_i^T\gamma_i}{(1+\gamma_i^T\gamma_i)^2}\left(\tilde{W}_i^T\gamma_i - \frac{1}{4}\tilde{W}_i^T\Upsilon_i^T\tilde{W}_i - \epsilon_{r,i}\right) \\ &+ k_{i2}z_i\left(F(z_i) + G_i u_i^{o,*}\right) + \frac{1}{2}z_i^T\Xi_i\nabla\epsilon_i\end{aligned} \tag{50}$$

Under the **Assumption** 4, we have

$$\begin{aligned}\dot{L}_{2i} \leq& -k_{i2}z_i^T\Pi_i z_i - k_{i1}\alpha_i||\tilde{W}_i||^4 + k_{i1}\varsigma_i \\ &+ \frac{1}{2}k_{i2}z_i^T\Xi_i\nabla\epsilon_i \\ \leq& -\frac{3}{4}\lambda_{\min}(\Pi_i)||z_i||^2 - k_{i1}\alpha_i||\tilde{W}_i||^4 \\ &+ \frac{\lambda_{\max}^2(\Xi_i)}{4\lambda_{\min(\Pi_i)}}\nabla\bar{\epsilon}_i + k_{i1}\varsigma_i\end{aligned} \tag{51}$$

In order to $\dot{L}_{i2} \leq 0$m the following conditions should be satisfied

$$\begin{aligned}&\alpha_i > 0 \\ &||\tilde{W}|| > \sqrt[4]{\frac{\varsigma_i}{\alpha_i} + \frac{\lambda_{\max}^2(\Xi_i)}{4k_{i1}\alpha_i\lambda_{\min(\Pi_i)}}\nabla\bar{\epsilon}_i}\end{aligned} \tag{52}$$

As analyzed for both cases, $\dot{L}_{2i} < 0$ is guaranteed with the conditions in (49) & (52). Based on the Lyapunov stability, the tracking error z_i and the estimated weight error $\tilde{W}_i$ are ultimate uniform bounded (UUB) regardless of whether the initial stabilizing control.

4 Simulation Result

In this session, numerous flight simulations were conducted using MATLAB R2022a on a personal laptop equipped with an Intel(R) Core(TM) i5-1145G7 CPU, 16 GB RAM, running Windows 11 64-bit. To demonstrate the tracking performance of the proposed control scheme, two controllers are considered for the comparison, including the controller in [23] and the proposed controller. The

Table 1 The parameter of the ADP-based controller

Controller	Parameters	Values
Position controller	$[Q_p, R_p, k_{p1}, k_{p2}]$	$[5I_6, I_3, 0.5, 0.2]$
Attitude controller	$[Q_\eta, R_\eta, k_{\eta 1}, k_{\eta 2}]$	$[10I_6, I_3, 0.5, 0.2]$

parameter of the quadrotor vehicle is configured as m=1.7 (kg), $J = \mathrm{diag}([4.7, 4.7, 9.8]) \times 10^{-2}$ (kgm^2), g=9.8 (m/s^2), l=0.2 (m), and $k_z = k_\tau = 1$. The initial condition of the quadrotor is set as $p(0) = [-2.0, 0.0, 1.0]^T$ (m), $\dot{p} = [0.0, 0.0, 0.0]^T$ (m/s), $\eta = [0.0, 0.0, 0.0]^T$ (rad), and $\dot{\eta} = [0.0, 0.0, 0.0]^T$ (rad/s). The desired position trajectory is chosen as

$$p_d = [2.0\sin(0.3t), 2.0\cos(0.3t), 0.5t]^T \tag{53}$$

The external disturbances acting on the quadrotor during flight are assumed to be

$$\begin{aligned} d_p &= 0.2\begin{bmatrix}\sin(1.0+0.7t)\cos(0.5t)\\ \sin(1.7+0.8t)\cos(0.6t)\\ \sin(0.5+0.4t)\cos(0.8t)\end{bmatrix}\\ d_\eta &= 0.2\begin{bmatrix}\sin(1.1+0.2t)\cos(0.3t)\\ \sin(1.4+0.3t)\cos(0.1t)\\ \sin(0.9+0.4t)\cos(0.1t)\end{bmatrix}\end{aligned} \tag{54}$$

The disturbance observer term is chosen as $\Lambda_p = \Lambda_\eta = 10I_3$. The optimal term is chosen as Table 1. The initial critic weights for position and attitude controllers are chosen as $W_p(0) = [5, 5, 5, 5, 5, 5, 5, 5, 5]^T$ and $W_\eta(0) = [1, 1, 1, 1, 1, 1, 1, 1, 1]^T$. The basic action function is chosen as

$$\begin{aligned} \varphi_p(z_p) &= [e_x^2, e_y^2, e_z^2, \dot{e}_x^2, \dot{e}_y^2, \dot{e}_z^2, e_x\dot{e}_x, e_y\dot{e}_y, e_z\dot{e}_z]^T\\ \varphi_\eta(z_\eta) &= [e_\phi^2, e_\theta^2, e_\psi^2, \dot{e}_\phi^2, \dot{e}_\theta^2, \dot{e}_\psi^2, e_\phi\dot{e}_\phi, e_\theta\dot{e}_\theta, e_\psi\dot{e}_\psi]^T\end{aligned} \tag{55}$$

To clarify the efficiency of the proposed control scheme, the simulation results are provided in Figs. 3, 4, 5, 6 and 7. The convergence of the disturbance estimation errors is shown in Fig. 5. As shown in the figure, the estimated errors converge to a small bounded neighborhood of zero after 0.1 seconds, reaching values as low as $|d_i| \le 0.01$ and $|d_j| \le 5 \times 10^{-3}$ with $i = x, y, z$ and $j = \phi, \theta, \psi$. Thus, external disturbances can be accurately estimated by using the nonlinear disturbance observer and subsequently compensated within the control loops. The convergence of the critic weights for position and attitude controllers is illustrated in Figs. 3 and 4, respectively. After 5 seconds, both critic weights converge and remain stable under the influence of external disturbance. The

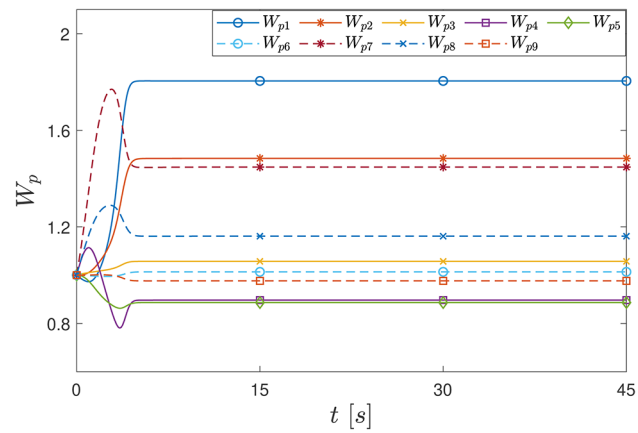

Fig. 3 The convergence of position weights

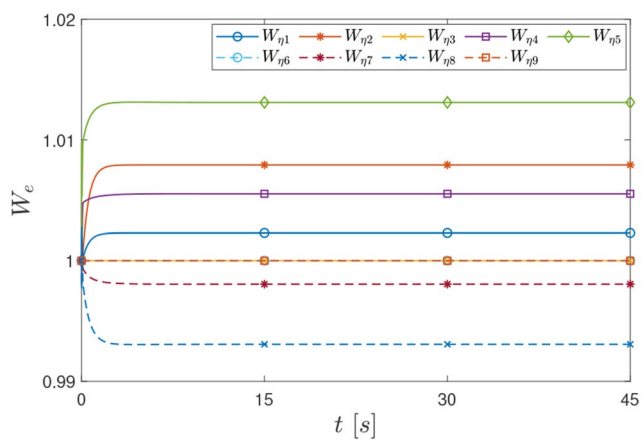

Fig. 4 The convergence of attitude weights

position and attitude responses of the quadrotor with the proposed controller are shown in Fig. 6. It is clearly observed that the quadrotor equipped with the proposed controller achieves excellent trajectory tracking performance. To further confirm the superior tracking performance of the quadrotor under the proposed controller, the reference trajectory and the actual flight trajectory in three-dimensional space are illustrated in Fig. 7. Furthermore, a comparison of the tracking error convergence performance between the proposed controller and the controller presented in [23] is provided in Figs. 8 and 9.

To analyze the tracking performance of the quadrotor under the different controllers, two performance indices in [28] are employed as follows:

– Integral of Squared Error (IAE)

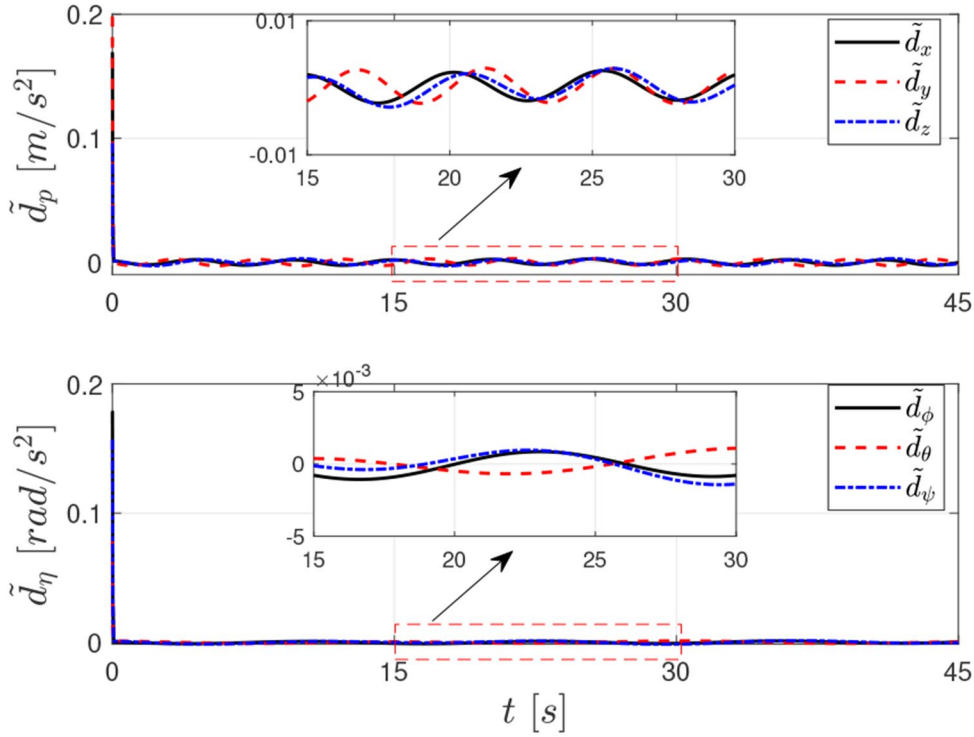

Fig. 5 The disturbance estimation errors

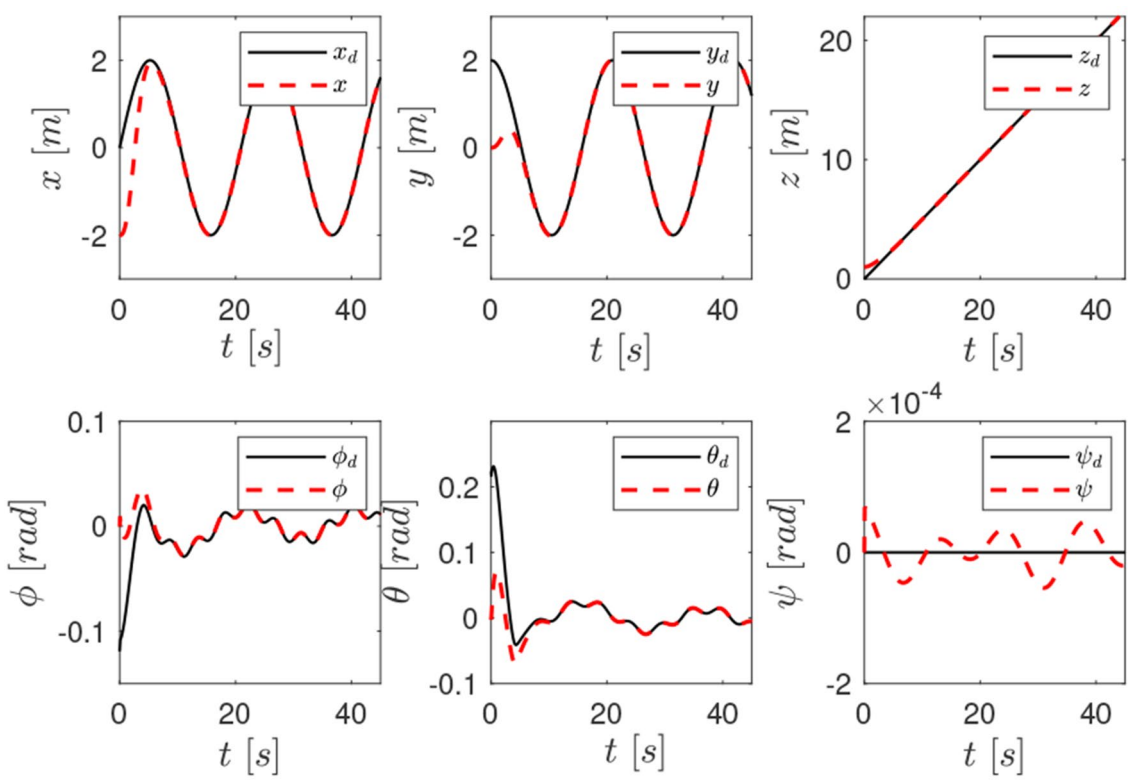

Fig. 6 The position and attitude response of the quadrotor

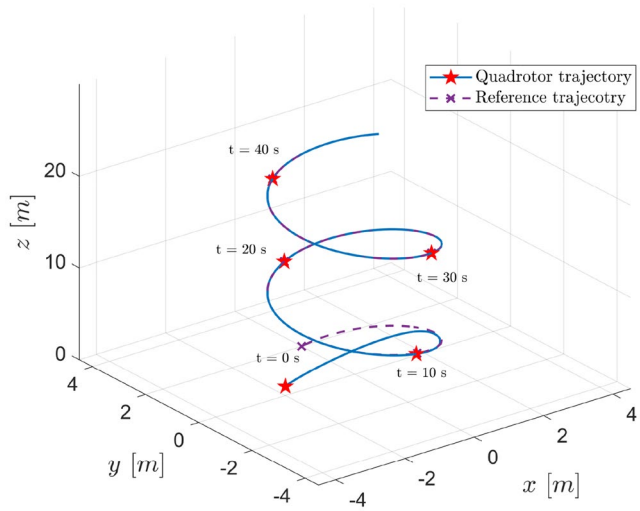

Fig. 7 The 3D trajectories of quadrotor and references

$$\mu^p_{IAE} = \int_0^T \left(|e_x(t)| + |e_y(t)| + |e_z(t)|\right)dt$$
$$\mu^\eta_{IAE} = \int_0^T \left(|e_\phi(t)| + |e_\theta(t)| + |e_\psi(t)|\right)dt \tag{56}$$

IAE (Integral of Absolute Error) was used to evaluate the overall magnitude of the deviation between the system output and the desired response.

– Integral of Time-weighted Absolute Error (ITAE)

$$\mu^p_{ITAE} = \int_0^T t\left(|e_x(t)| + |e_y(t)| + |e_z(t)|\right)dt$$
$$\mu^\eta_{ITAE} = \int_0^T t\left(|e_\phi(t)| + |e_\theta(t)| + |e_\psi(t)|\right)dt \tag{57}$$

ITAE (Integral of Time-weighted Absolute Error) was applied to emphasize the impact of errors that persist over time, highlighting the system's ability to quickly minimize deviations.

The tracking performance comparisons of the proposed controller and the compared controller in shown in Table 2. As shown in Figs. 8 and 9 and Table 2, we can see that the tracking performance of the proposed method is better than the method in [23]. Compared to the method in [23], the proposed approach achieves faster tracking error convergence and significantly reduces "chattering" and overshoot.

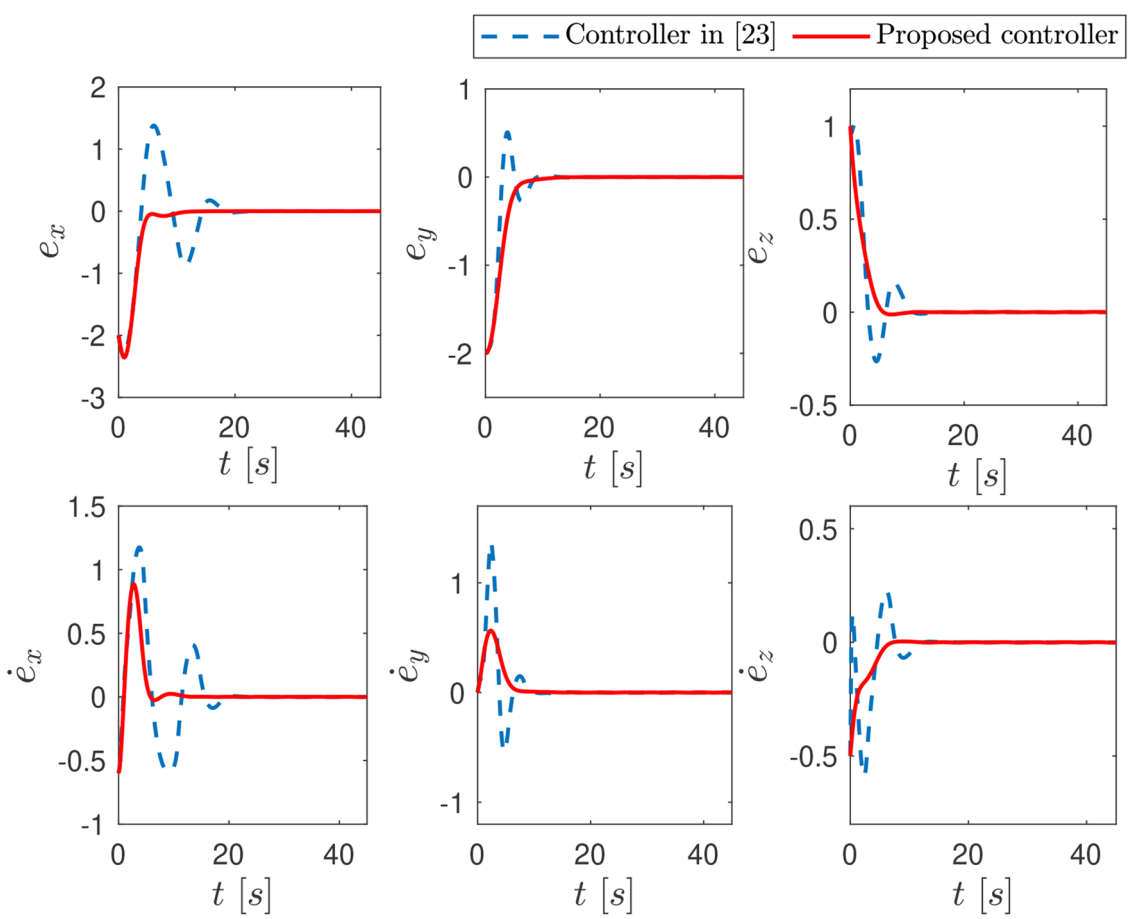

Fig. 8 The tracking errors of the position loop

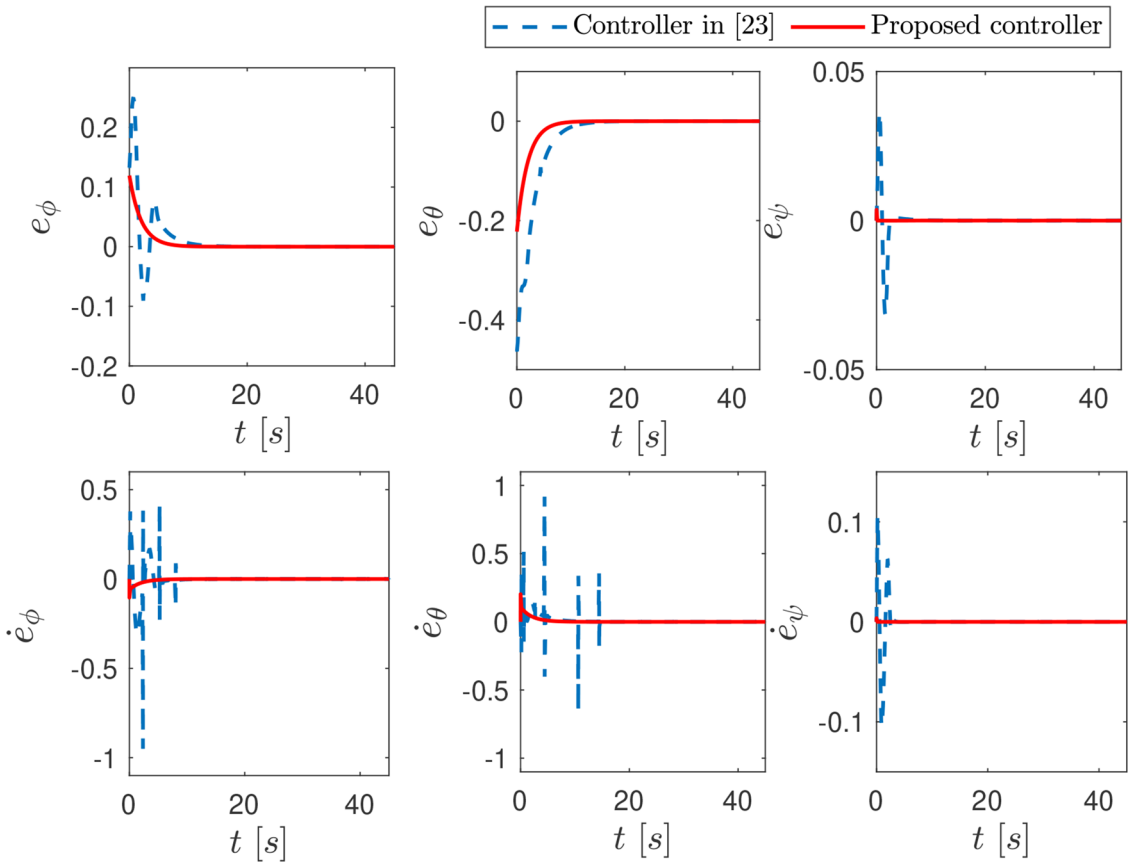

Fig. 9 The tracking errors of the attitude loop

Table 2 The tracking performance comparisons

Controller	μ^p_{IAE}	μ^η_{IAE}	μ^p_{ITAE}	μ^η_{ITAE}
Controller in [23]	22.80	2.02	9.98×10^4	5.54×10^3
Proposed controller	15.20 ($\downarrow$ 33%)	0.69 ($\downarrow$ 66%)	$3.11 \times 10^4 \downarrow$ (69%)	$1.45 \times 10^3 (\downarrow$ 74%)

Theoretically, the "chattering" issue in the compared control scheme is caused by using the discontinuous function $sign(.)$ in the terminal sliding mode observer. In contrast, the superiority of our method lies in its ability to compute a smooth optimal control policy even under the influence of disturbances, while effectively mitigating the chattering effect.

5 Conclusion

This paper invests in designing a robust tracking controller for quadrotor UAVs subjected to unknown external disturbances. The problem of achieving optimal trajectory tracking is not usually discussed in conventional control approaches for quadrotor UAVs. To overcome this limitation, this work proposes a robust optimal control scheme, consisting of an ADP-based controller and a nonlinear disturbance observer. The advantage of this work is to achieve a near-optimal tracking objective for the quadrotor system under the influence of unknown external disturbances through the NN-based update law of the online critic. The quadrotor system with the proposed controller is ultimately uniformly bounded (UUB) stability. A simulation result is provided to illustrate the effectiveness of the proposed method. In future work, we plan to consider practical constraints such as input saturation and state limitations (e.g., roll, pitch, and yaw angle bounds), which would further enhance the applicability of the proposed control scheme in real-world scenarios.

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Data Availability The data that supports the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethical Approval Not applicable.

Competing Interests The authors declare that they have no competing interests.

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